Element B

Prior Examples:

#	Prior Designs	Pros	Cons
1	Interactive Floor and Wall Panels	- When kids step on the different tiles they change color and make sounds - Panels are very durable, kids can hit, jump, and roll a wheelchair on top of the panel and it won't break - Kids can play games on the panel that test memory and challenge hand eye coordination - Offers three different size panels (9 panels, 12 panels, and 16 panels) -Panels can be mounted anywhere so kids can interact with the panels while standing, sitting, and laying	- Kids with very limited mobility will not be able to access the entire panel (smallest size is 45.7" x 40.9" x 1.6" deep) -If the area is not dark the lights will not be as bright making it less stimulating for the kids
2	Fibre Optic Light Plume	- Completely safe to touch and play with - Changes color to provide stimulation and can reflect user's mood/preferences - Small dimensions (8 cm width x 34.5 cm height x 8 cm depth), which provides easy mobility and transportability - Low cost and relatively easy to construct - Creates a calming environment and helps to calm down overstimulated kids - The light strands can be touched and moved around, providing some interactivity	- Only provides static lighting, no flashing or moving - Limited interactive elements - Could easily be damaged because of the numerous extruding parts
3	LED Optic Projector	- It can display a wide array of moving colors and shapes - This as a design would be able to cater to multiple preferences in color or shapes -It is a passive form of stimulation which could be good	-It doesn't allow for any other kinds of stimulation(smell or touch) -It way be hard to find a good place to put the projector so that the image is not obscured

		depending on the child.	
4	LED Infinity Mirror	- Hangs on wall which keeps floor space free and open - Light color can be manually changed or programmed to change autonomously - Provides visual stimulation which can either be calming or intriguing for kids - Allows kids to learn about how mirrors work if that's something important to the teachers - The infinite illusion creates a thing for the kids to get lost in	- Very expensive and difficult to build - Mirror could easily be scratched and damaged by kids - Would have to be very well secured to the wall to prevent injury to kids or damage to the mirror - Only stimulates the visual sense
5	Light Spinning Wand	- Thick ergonomic handle - Cause and effect - Colorful visuals - Vibrant Colors - Small (7.5" long) and maneuverable, allowing for ease of transportation and ease of use for children of all sizes - Highly interactive device that can be used to stimulate children and develop senses - Takes batteries, therefore doesn't need to be plugged in, making it much more portable	- Low durability, not designed for rough use and will likely break if dropped on a hard surface - May become expensive to constantly replace batteries - Could be overstimulating to certain children

Analysis of Prior Examples:

Interactive Floor and Wall Panels:

The interactive floor and wall panels are a unique way to calm kids down that are overstimulated. The panels allow the kids to get moving compared to other light sensory devices where the kids can only watch. It also offers multiple games/levels to keep kids engaged and wanting to use it again. On top of the panels changing color when touched they also can make sounds. The panels come in different sizes to be able fit in big and small spaces. The panels are extremely durable and can handle heavy jumping, hitting, and a wheelchair being rolled over them. The panels can be mounted on the floor and walls allowing for kids to be able to interact with the panels standing, sitting, and laying. On the downside if you install the panels in a bright room it won't look as bright making it less stimulating for the kids. Even though it can be used by kids in wheelchairs, kids with limited mobility won't get the most out of the device. Overall interactive planes are a great way for kids to relax.

Fibre Optic Light Plume:

The Fibre Optic Light Plume is an effective sensory device when it comes to providing a calming experience for kids who may feel overstimulated. Its color-changing capabilities combined with its unique shape and structure provide an intriguing device that kids can use to calm themselves down. Each light strand can be touched and played with, providing some interactivity to the device. This can provide touch sensory along with visual sensory. The color can be changed to reflect the user's mood or current preference, which further contributes to calming them down. Additionally, the plume is completely safe to touch and mess with, making it easy to put in a sensory room without risk of injury. However, the lamp could easily be damaged with too much use because of its numerous extruding parts, giving it a potentially short lifespan. Also, the lights don't flash or move at all, which could either be a positive or negative depending on who is using it. Overall, the fibre optic light plume is a good middle-of-the-road sensory device. It combines visual and touch sensory to create a device that children can interact with and touch, or just sit and admire depending on what their current circumstances are.

LED Optic Projector

The LED Optic Projector is effective in providing visual stimulation. The device can project over 200 different designs, shapes, patterns, and moving displays, and furthermore can be aimed at any point in a room. The designs can cover ceilings, walls, floors, and create a beautiful design and stimulating environment wherever it is located. Generally, the projector is proficient in accommodating passive visual stimulation needs. Equipped with a manual timer, the device can be programmed to activate on a timed schedule at the teachers discretion. Additionally, the projector is further equipped with a table and wall mount, meaning that the device can be placed and moved to any part of a room. Besides the array of visual displays, the projector does not provide any further interactive stimulation or other intricate functions. It is simply utilized for visual sensory displays.

LED Infinity Mirror:

The LED Infinity Mirror is effective at providing purely visual stimulation, at the cost of almost no interactivity. It's a device that the user can get lost in and mesmerized by, allowing them to calm down if overstimulated or just zone out if needed. Also, its colors can be manually changed, or they can be coded to automatically change. This gives more control to the user, and allows them to control the space they're in. It's hung on the wall, which keeps the floor space of the sensory room open and available for movement and additional devices. However, it's very expensive and requires a lot of complicated parts and coding during construction. It could also be very easily damaged through scratching and prolonged touching, or it could fall off the wall and cause injury to the kids or damage to the mirror. Additionally, it's lacking in interactivity as it only stimulates the visual sense and doesn't provide any other kinds of sensory stimulation. This could be seen as a pro or a con depending on the child and what it's meant to be used for. Overall, the LED infinity mirror is an effective sensory device for children who need to be calmed down or just zone out.

Light Spinning Wand:

The spinning wand is a device that will display an array of light that gets created from a light source being spun at rapid speeds. It provides visual stimulation while giving a cause and effect stimulation. It is a simple product and the concept can be implemented in other design ideas. The cons of this product is that the rapid spinning of light could be unsettling for some users. It is noted that durability is a point of concern. The "crystal ball" on top of the device could crack if dropped on a hard surface, and the chance of it being dropped or thrown when in the possession of young children is high. Some safe sides of the product is that the large handle and ball eliminates the choking hazard. In summary, The spinning wand requires an interest in interactive visual stimulation and a tolerance for rapidly spinning lights.